

Paint Rock spectral–biodiversity relationships

Main question: can hyperspectral spectral heterogeneity serve as a proxy for biodiversity and phylogenetic diversity in the Paint Rock Forest Dynamics Plot?

The analysis evaluates whether quadrat–level hyperspectral heterogeneity tracks biodiversity and phylogenetic diversity across 10 m, 20 m, and 50 m grain sizes.

SA entropy mean is treated as the primary response. For most quadrats it is the mean of 70 bootstrap iterations using up to 5,000 retained pixels; the small edge quadrats use fewer pixels.

Models use standardized predictors so coefficients are comparable within scale. Documented 10 m and 20 m edge quadrats are excluded from primary inferences.

Primary response: SA entropy mean (`spec_sa`). For most quadrats it is the mean of 70 bootstrap iterations using up to 5,000 retained pixels; the small edge quadrats use fewer pixels.

Outputs in this report are descriptive and inferential OLS summaries with spatial residual checks. When residual Moran's I remains significant, p–values should be interpreted with caution.

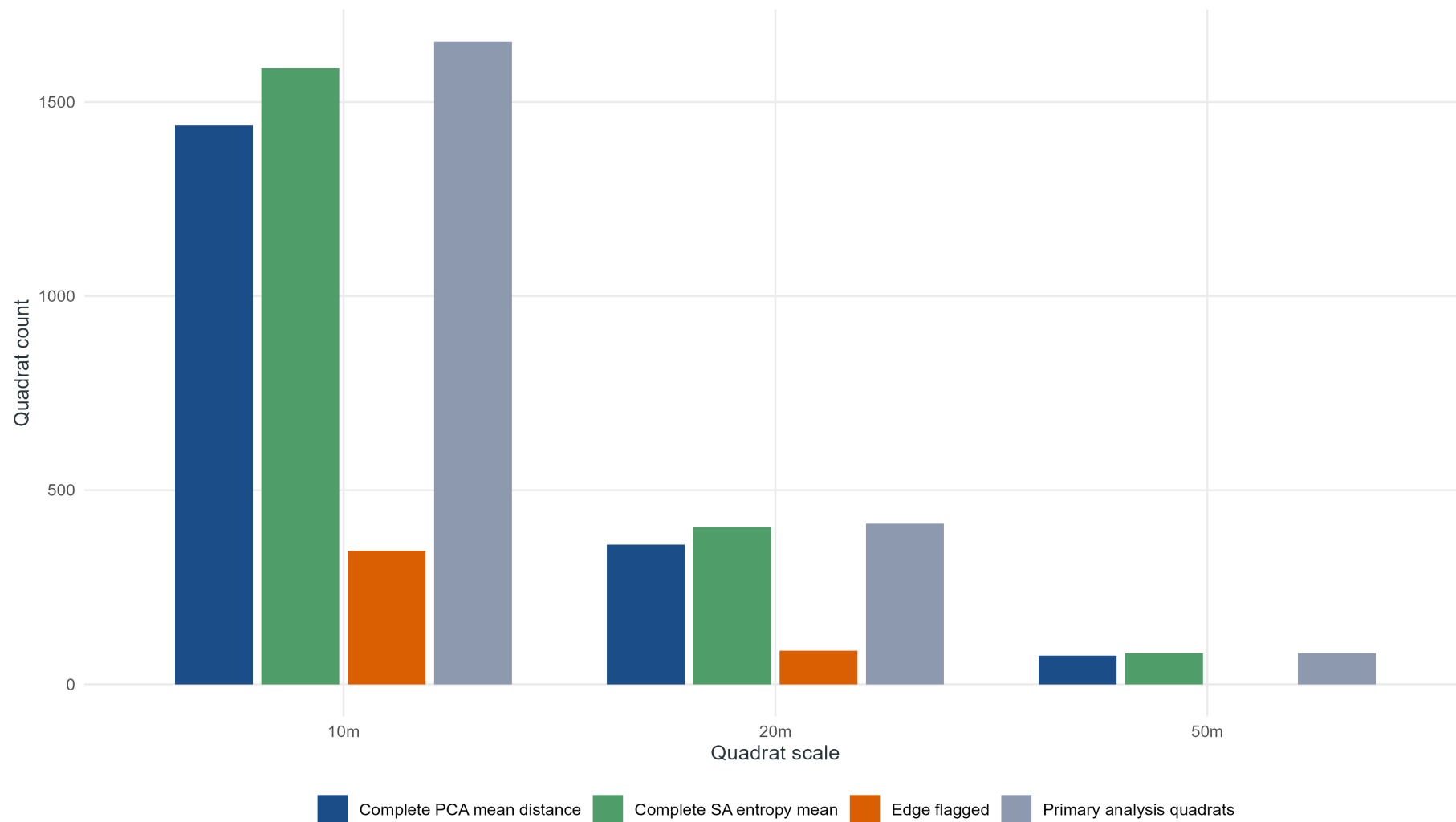
Analysis sample sizes

Scale	All quadrats	Primary n	Complete SA	Complete PCA mean	Edge flagged
10m	2000	1656	1587	1440	344
20m	500	414	405	360	86
50m	80	80	80	74	0

Coverage and data completeness

Analysis coverage by scale

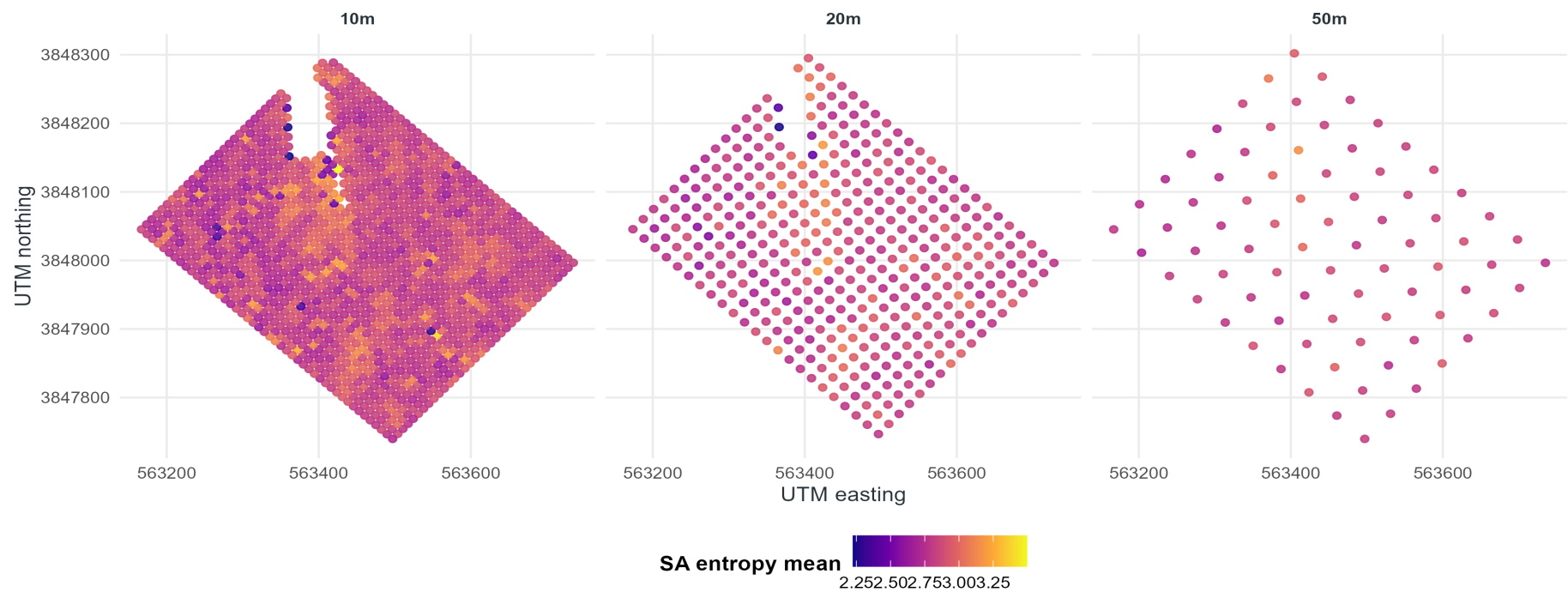
Primary analysis excludes documented 10 m and 20 m edge quadrats; 50 m has no separate edge rule.



Spatial pattern of primary spectral heterogeneity

Spatial pattern of SA entropy means

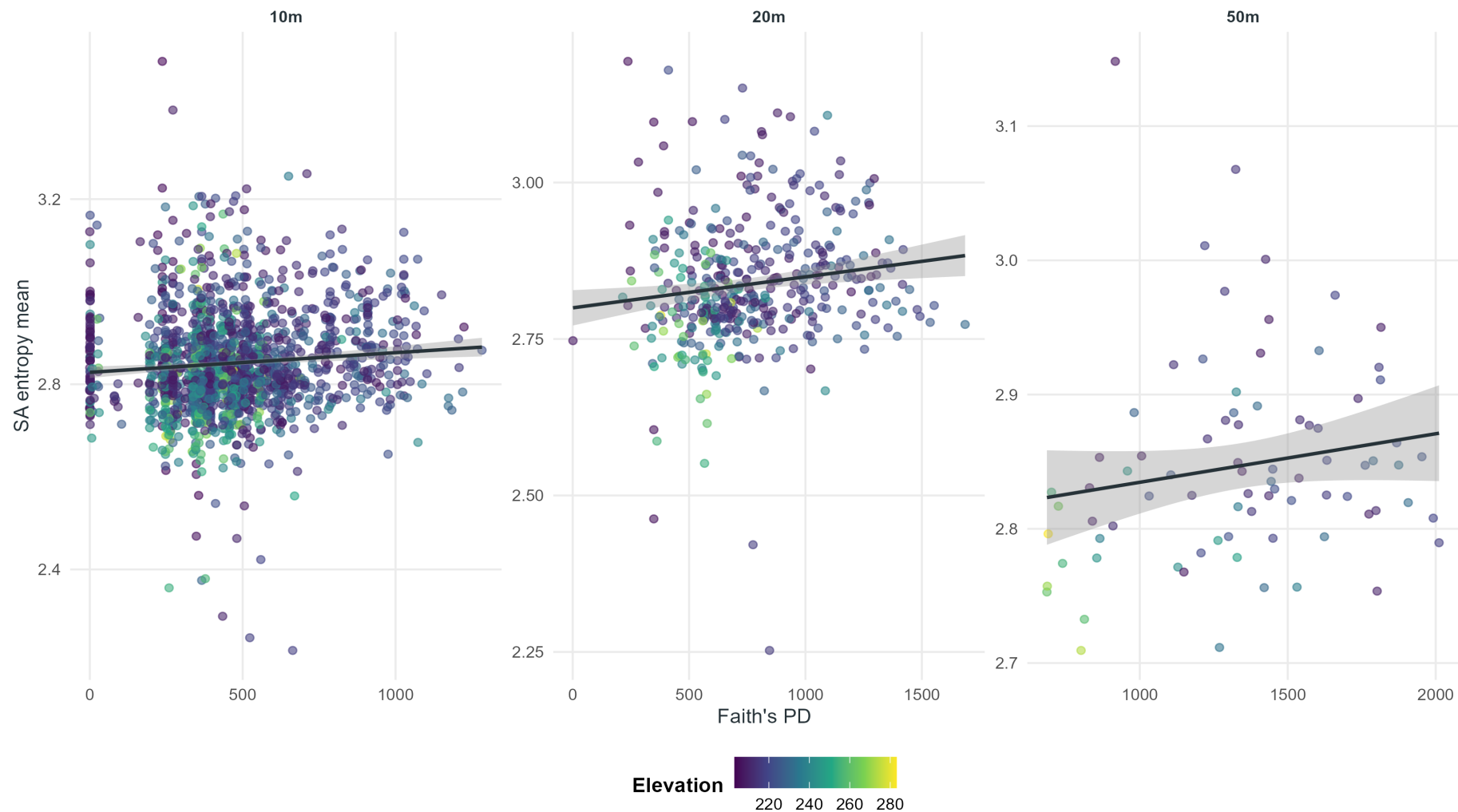
Most SA values are means of 70 bootstrap iterations; centroid maps use plant-diversity quadrat centroids in NAD83 / UTM zone 16N.



SA entropy mean versus Faith's PD

SA entropy mean versus Faith's phylogenetic diversity

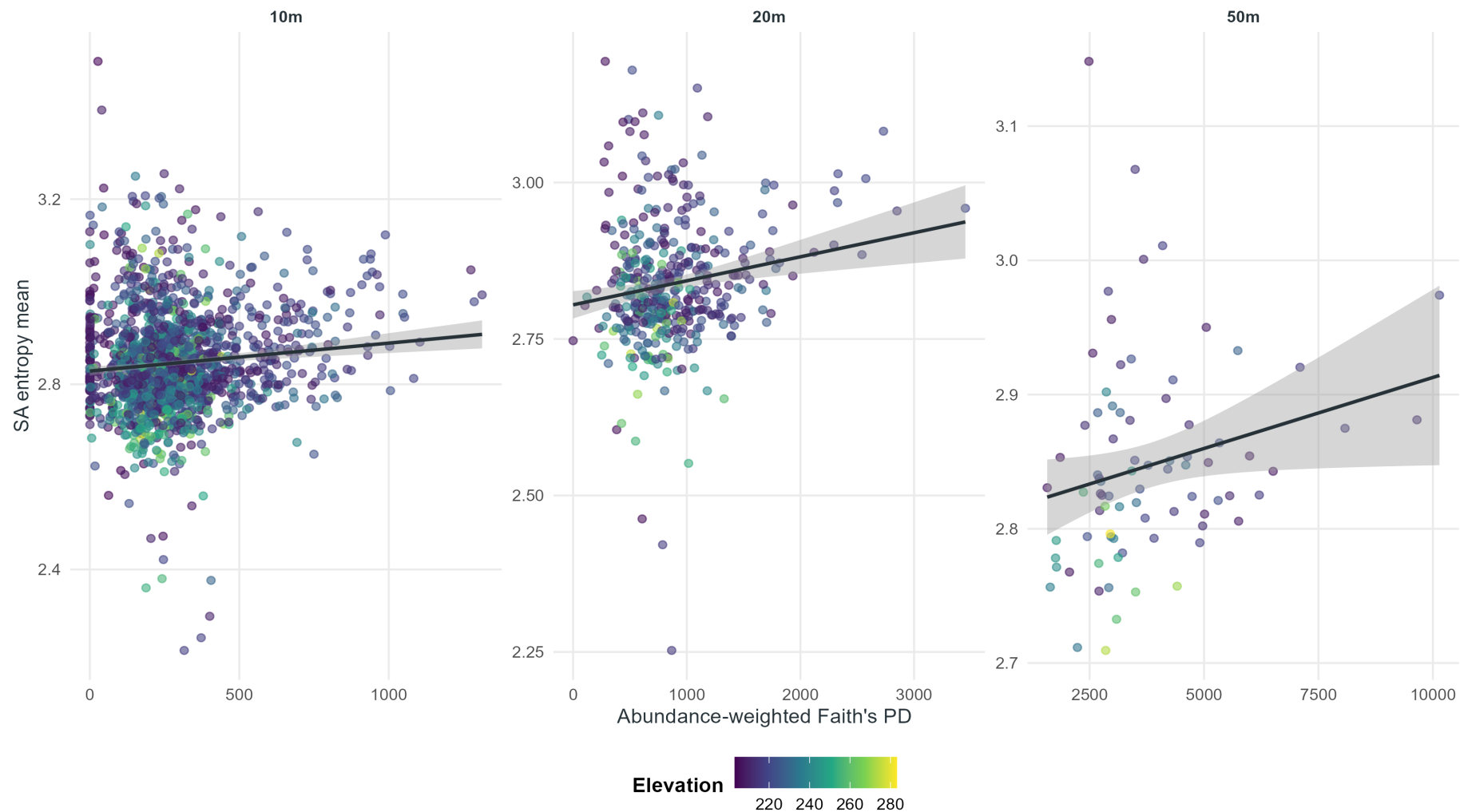
Lines show simple linear fits within scale; most SA values are bootstrap means, with exact entropy only for small quadrats.



SA entropy mean versus abundance-weighted Faith's PD

SA entropy mean versus abundance-weighted Faith's PD

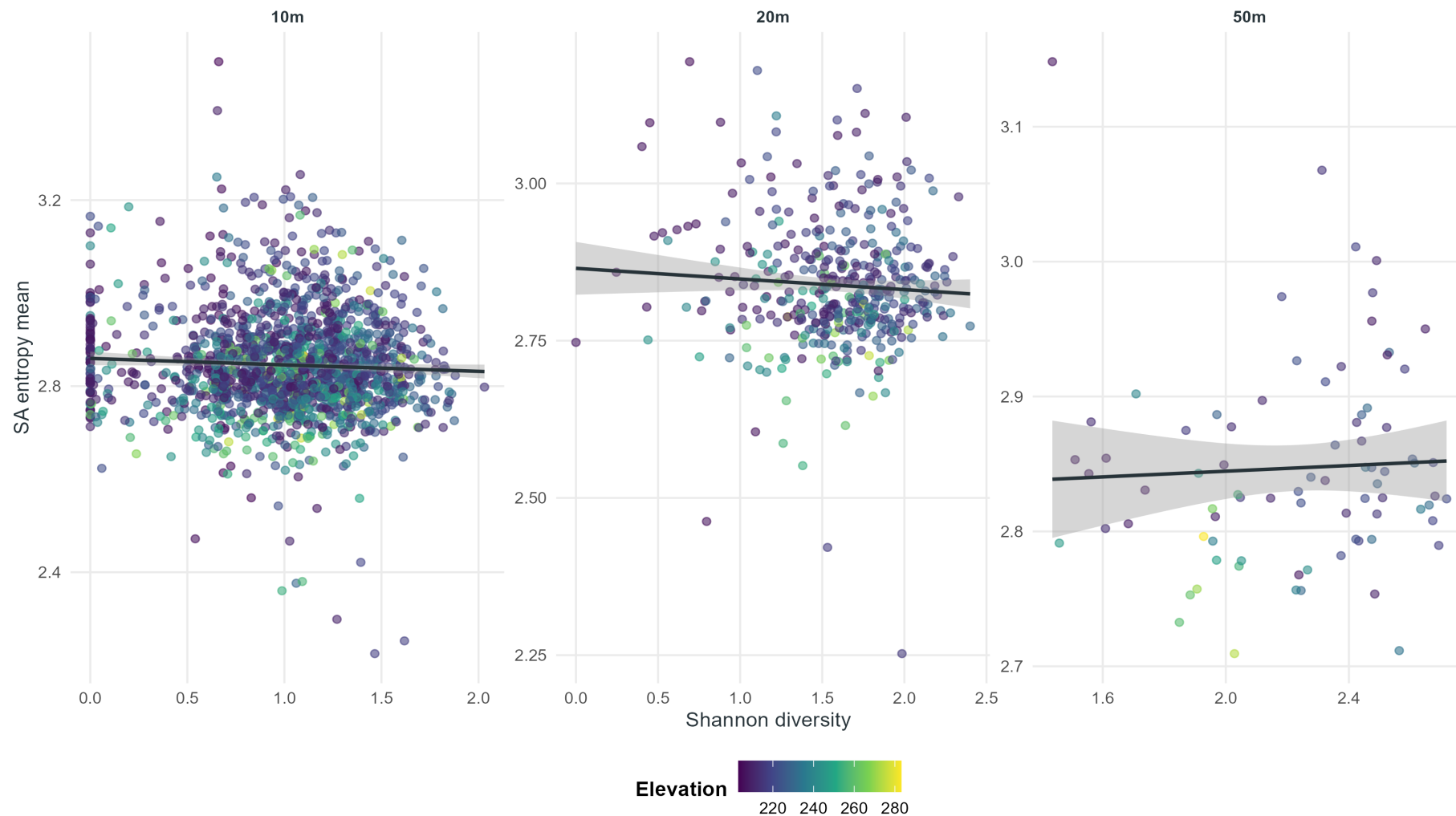
Lines show simple linear fits within scale; most SA values are bootstrap means, with exact entropy only for small quadrats.



SA entropy mean versus Shannon diversity

SA entropy mean versus Shannon diversity

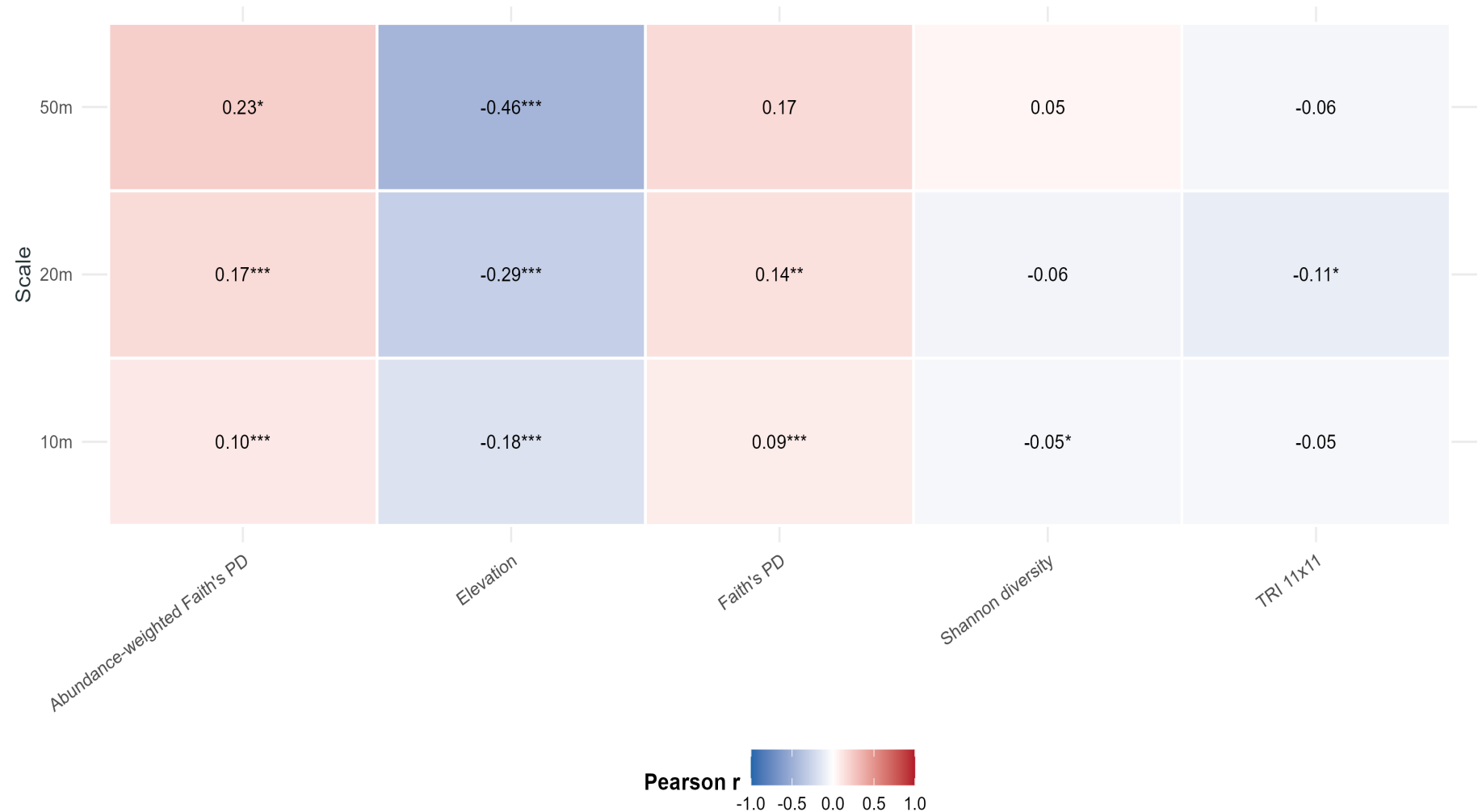
Lines show simple linear fits within scale; most SA values are bootstrap means, with exact entropy only for small quadrats.



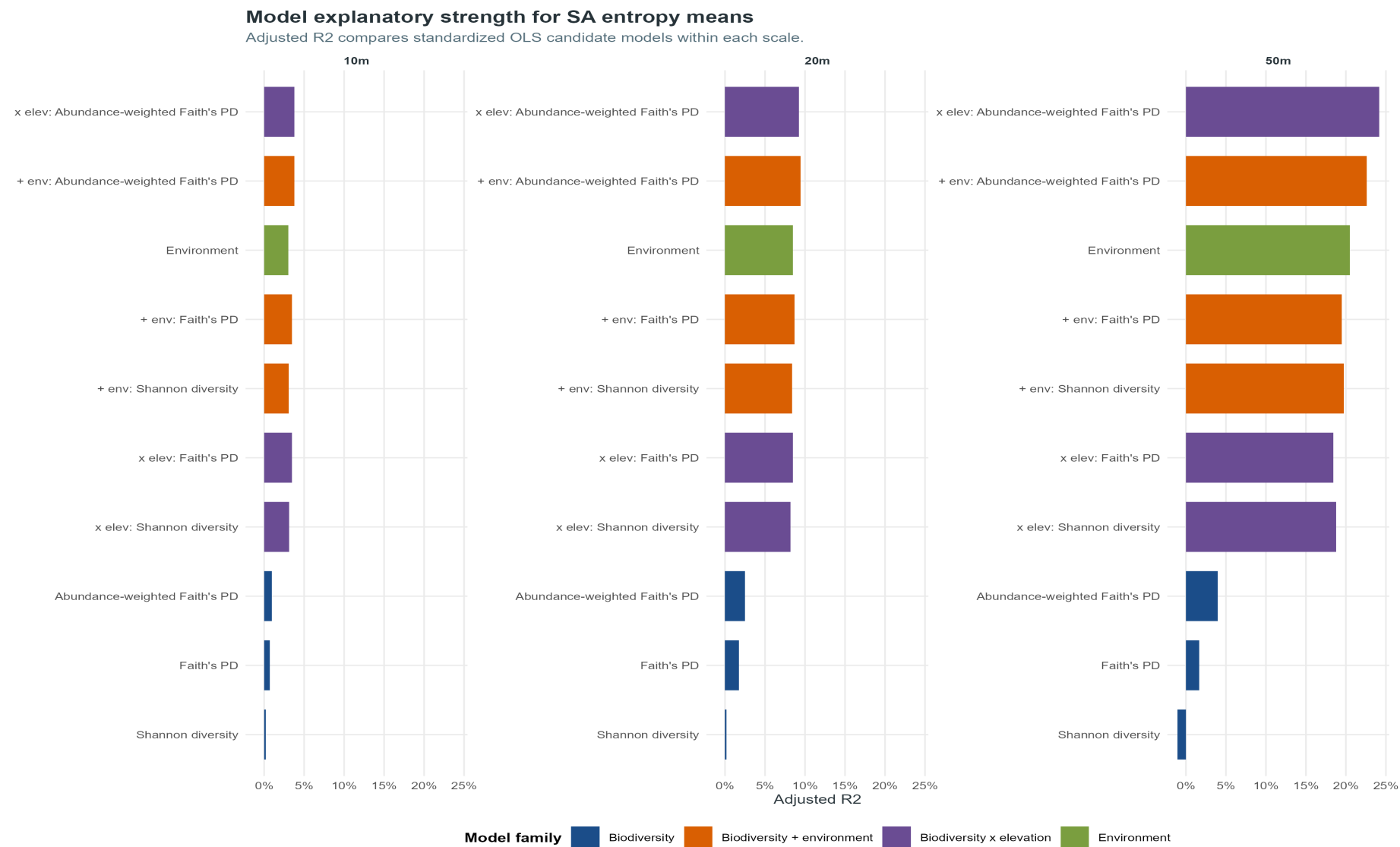
Primary correlations

Primary correlations with SA entropy means

Pearson r values for primary-analysis quadrats. Asterisks mark $p < 0.05$, 0.01 , and 0.001 .



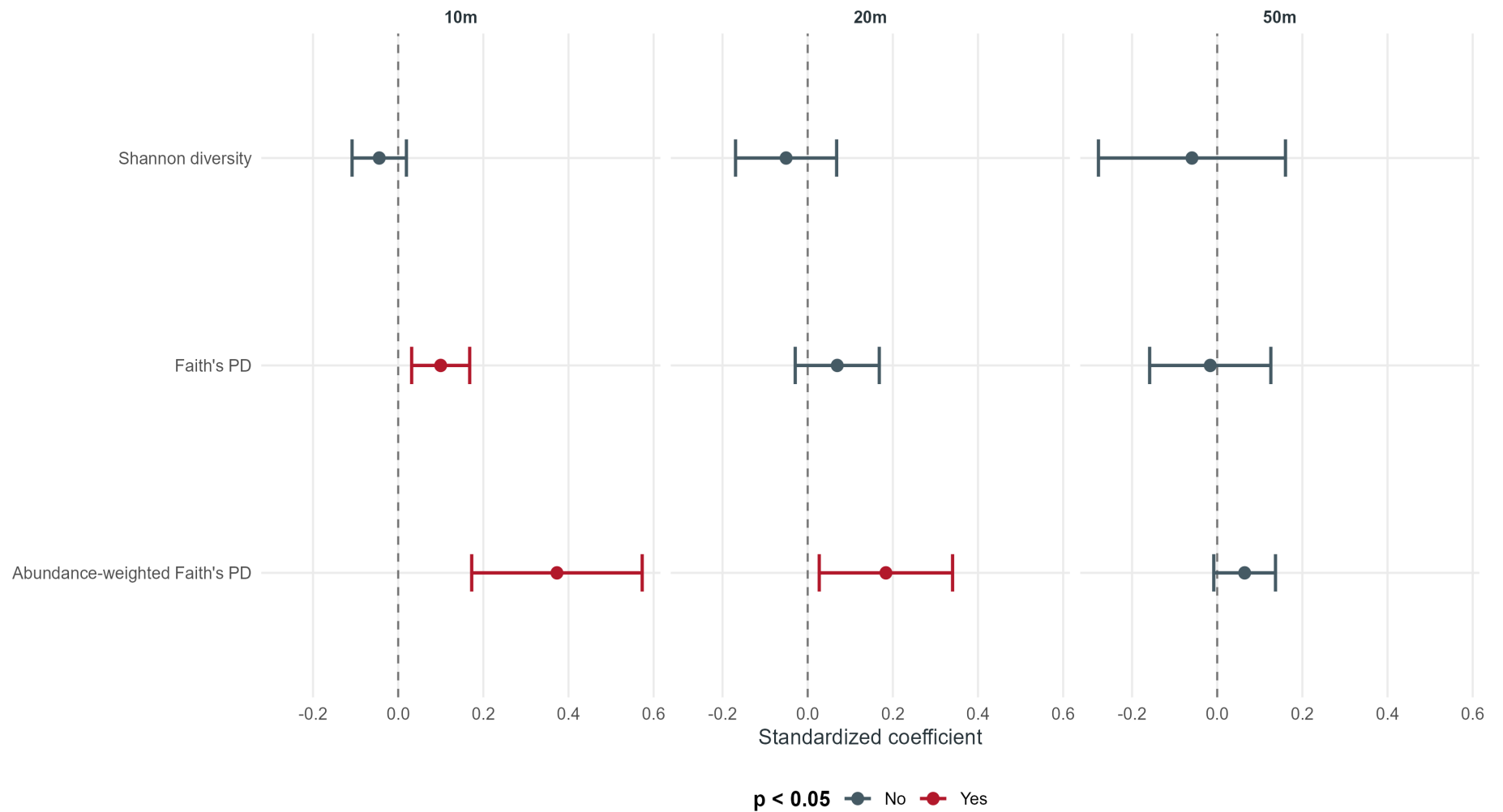
Model comparison



Biodiversity effects after environmental controls

Standardized biodiversity effects after environmental controls

Points are standardized coefficients; bars are 95 percent confidence intervals.



Residual spatial autocorrelation

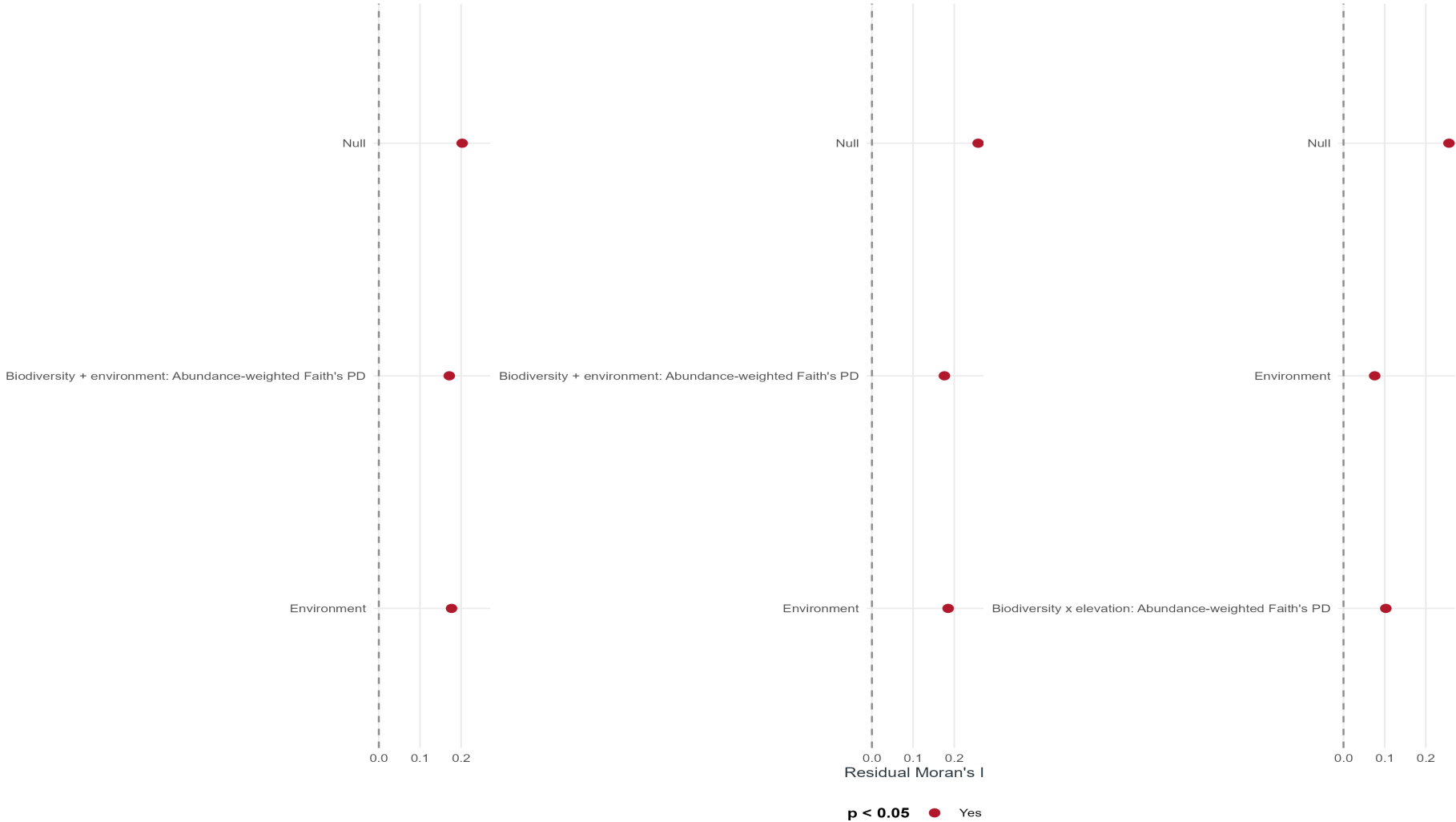
Spatial autocorrelation in model residuals

Moran's I is calculated from 8-nearest-neighbor residuals. Significant residual structure cautions against over-interpreting p-values.

10m

20m

50m



Best-supported primary-response model by scale

Scale	Model	n	Adj. R2	AIC	Delta AIC
10m	Biodiversity + environment: Abundance-weighted Faith's PD	1587	0.038	4604.5	0.0
20m	Biodiversity + environment: Abundance-weighted Faith's PD	405	0.094	1086.2	0.0
50m	Biodiversity x elevation: Abundance-weighted Faith's PD	80	0.242	159.4	0.0

Primary biodiversity correlations

Scale	Predictor	n	Pearson r	Spearman r	p
10m	Faith's PD	1587	0.088	0.106	<0.001
10m	Abundance-weighted Faith's PD	1587	0.102	0.080	<0.001
10m	Shannon diversity	1587	-0.051	-0.019	0.043
20m	Faith's PD	405	0.140	0.211	0.005
20m	Abundance-weighted Faith's PD	405	0.166	0.124	<0.001
20m	Shannon diversity	405	-0.065	-0.020	0.195
50m	Faith's PD	80	0.171	0.245	0.130
50m	Abundance-weighted Faith's PD	80	0.228	0.281	0.042
50m	Shannon diversity	80	0.047	0.139	0.678

Spatial residual diagnostic summary

Scale	Model	n	Moran I	p
10m	Null	1587	0.203	<0.001
10m	Environment	1587	0.177	<0.001
10m	Biodiversity + environment: Abundance-weighted Faith's PD	1587	0.171	<0.001
20m	Null	405	0.258	<0.001
20m	Environment	405	0.185	<0.001
20m	Biodiversity + environment: Abundance-weighted Faith's PD	405	0.176	<0.001
50m	Null	80	0.256	<0.001
50m	Biodiversity x elevation: Abundance-weighted Faith's PD	80	0.103	0.012
50m	Environment	80	0.076	0.042

Interpretation guardrails

1. A positive biodiversity coefficient supports the spectral–variation hypothesis only when its confidence interval is mostly above zero and residual spatial autocorrelation is low.
2. Environmental controls are included because topography can affect both canopy composition and observed reflectance. Models that retain biodiversity controls are more likely to be robust.
3. The 50 m scale is useful for scale dependence, but sample size is small. Large coefficients at 50 m should be read alongside confidence intervals and residual spatial autocorrelation.
4. Known spectral missingness and PCA exclusion rules are preserved. No spectral values were imputed.
5. ``spec_sa`` is usually the mean of 70 bootstrap iterations using up to 5,000 retained pixels; only the small exact subset used all retained pixel pairs. Bootstrap iterations are not independent.